

Quiz 1 — Practice

STAT 184 — Summer 2026

Name: _____

Date: _____

About this practice quiz

- This is a **practice** quiz to help you prepare for Quiz 1.
- It mirrors the **format, length, and topic distribution** of the real quiz.
- The actual quiz will have different specific questions, but they will test the same concepts.
- **Recommended setup:** Give yourself 20 minutes, closed-book, no electronics. Then check your answers against the key.
- 9 questions, 30 points total.

1. (2 pts) Multiple choice — assignment

Which of the following is the **preferred** way (in this course) to assign the value 100 to an object named `score`?

- (A) `score = 100`
- (B) `score == 100`
- (C) `score <- 100`
- (D) `100 <- score`
- (E) `assign(score, 100)`

Answer: _____

2. (2 pts) Multiple choice — packages

You just opened RStudio for the day and want to use functions from the `ggplot2` package. You installed `ggplot2` a few weeks ago. Which command do you need to run to make the functions available in your current session?

- (A) Reinstall the package with `install.packages("ggplot2")`.
- (B) Load it with `library(ggplot2)`.
- (C) Nothing — once installed, it's loaded automatically every session.
- (D) Restart R and try again.
- (E) Run `update.packages("ggplot2")` first.

Answer: _____

3. (4 pts) Code reading — predict the output

For each of the following, write what R prints to the console.

(a) `sqrt(144)`

(b) `c(10, 20, 30) + 5`

(c) `15 %% 4`

(d) `mean(c(1, 3, 5, 7, 9))`

4. (4 pts) Code reading — identify the parts

Consider the following code:

```
Movies |>
  filter(year == 2010) |>
  summarise(avg_rating = mean(rating))
```

(a) Name the **data table**:

(b) Name **two functions** used:

(c) Name **one variable** (column):

(d) Name **one constant**:

5. (3 pts) Multiple choice — data structures

Which of the following best describes a **vector** in R?

- (A) A one-dimensional collection of values, all of the same type.
- (B) A two-dimensional collection of values, all of the same type.
- (C) A two-dimensional collection where each column can hold a different type.
- (D) An ordered list of unrelated R objects, possibly of different types.
- (E) A single number stored in an object.

Answer: _____

6. (3 pts) Code reading — function arguments

Consider this call:

```
round(3.14159, digits = 3)
```

(a) What is the **name of the function** being called?

(b) How many **arguments** are passed to it?

(c) Is the first argument (3.14159) **named** or **positional**?

7. (3 pts) Short answer — comments on code

In one or two sentences, explain the difference between running `mean(c(2, 4, 6))` and running `m <- mean(c(2, 4, 6))`. Which one stores the result for later use?

8. (4 pts) Code reading — histograms

Consider this code:

```
ggplot(penguins, aes(x = bill_length_mm)) +  
  geom_histogram(binwidth = 2, fill = "steelblue", color = "white")
```

(a) Which variable will be on the **x-axis** of the plot?

(b) How wide is each bar (bin) in this histogram?

(c) What color will the **inside** of the bars be?

(d) What color will the **outline** of the bars be?

9. (5 pts) Short answer — debug the code

A classmate runs this code and gets an error:

```
ggplot(penguins, aes(x = body_mass_g)) %>%  
  geom_histogram()
```

(a) Identify **what's wrong** with this code. (Hint: think about how `ggplot2` layers are joined.)

(b) Write the **corrected** version of the code.

End of practice quiz — Total: 30 points

Check your answers using the answer key.